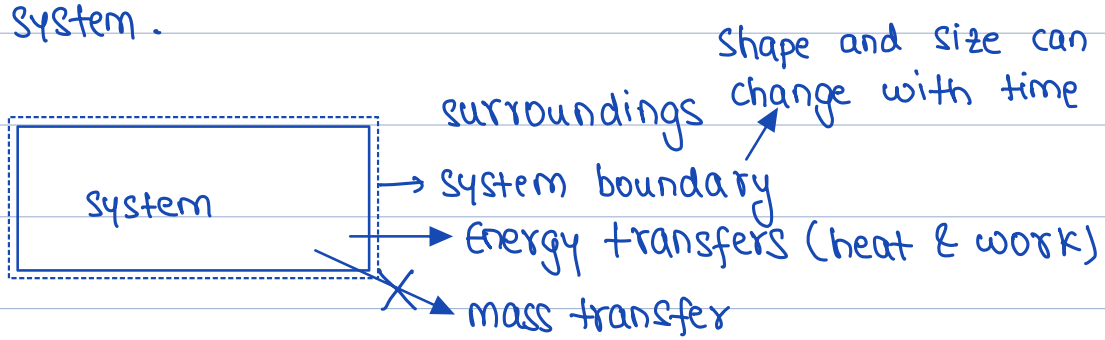
Fundamentals:

- Steam engines & Internal combustion engines convert heat to work.
- Thermofluids need cooling (Chip carrier, electrical equipment cooled by fan) failure rate increases with increase in temperature.
- All processes must obey laws of thermodynamics.
- Law of energy conservation: 1st law of thermodynamics. Energy can neither be created or destroyed, it can be converted from one form to another.
- Changes do not take place in any direction
- If a process happens to a system and neither of the pressure, volume, temperature change, there still could be a change in the surroundings
- 100% heat can't be converted into work (Kelvin-Plank)
- 100% can be converted into heat (Joule's experiment)
- Second law: directionality and puts limits on conversion of energy from one form to another.

- Macroscopic - average effects of several distinguishable constituents.
- System: Fixed mass and identity of matter on which attention is focussed.
- Surrounding: The rest of the universe, which lies outside the system.
-



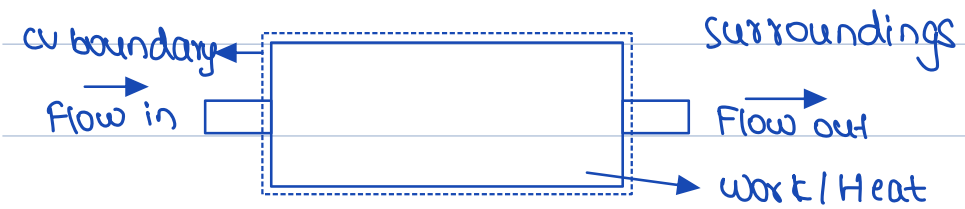
- Examples of system:

  a) electric bulb b) Gas in piston-cylinder device

  c) Electrical heater d) Photovoltaic solar cell

  e) domestic refrigerator f) balloon filled with gas

- Control volume: fixed volume in space on which the attention is focussed, energy & mass transfer can occur



There should atleast be an inflow or outflow

- Examples of CV:

  a) Water pump b) Air conditioner c) Car engine

- Property: Any measurable & observable macroscopic characteristic of a substance (mass, volume, ρ, T, θ, elevation, c, KE, H, S etc.

- Intensive property: independent of mass of system (ρ, T, ρ, θ)

- Extensive property: Dependent on size/mass of system (V, PE, KE)

- Specific property: Extensive property per mass (v, e, h, u)

- State: collection of all properties of a system when the system is in thermodynamic equilibrium
  - All property values throughout the system should be uniform
  - Atleast for a moment they should be invariant of time

- property is called point function / state variable

- Only 2 properties are required to define the thermodynamic coordinate system

- thermodynamic equilibrium is ensured by the attainment of mechanical, thermal and chemical equilibrium.



⇒ No talk of thermodynamic equilibrium (isolated system)

- Change of state occurs when any property of the system changes its value.
- Path is the locus of all intermediate states that the system passes through when there's a change of state
- Process is a sequence of states by which change of state takes place in a particular path.
- cyclic process: system changes its state more than once and finally reaches the same initial state.
- Quasistatic process: slow process in which the intermediate states are well accounted. This process has several intermediate equilibrium states.
- Properties are defined only in Quasistatic processes.
- To calculate energy transfer during change of state, the exact process or path followed by the system should be known

- Energy transfers can be calculated only when the process is quasistatic.
- Quasistatic process may also be a reversible process (ignoring dissipative forces)
- Properties are unique to a state and are path independent
- Properties are exact differentials.

$z = f(x,y)$ $\quad dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy \quad \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ → z is exact and state function

(dz: total differential; $M = \left(\frac{\partial z}{\partial x}\right)_y$, $N = \left(\frac{\partial z}{\partial y}\right)_x$)

- For a fluid at rest, the pressure exerted is same in all directions.
- Manometer & Bourdon gauge measure pressure w.r.t. surrounding pressure
- Barometer measures absolute pressure.

Work:

- An energy interaction between the system and surrounding
- Work is said to be done by a system, during a given operation, if the sole effect, external to the system could be reduced to the raising of weight through a dist.

- $W_{\text{by the system}} > 0$ $W_{\text{on the system}} < 0$
- If all the components are included in the system, there is no work interaction
- Process carried out by the system must be resisted by the surroundings for work interaction.
- Work-energy theorm: The net work done by forces on an object equals change in its kinetic energy.
- Displacement work:

$P = P_{ambient} + \Delta P$



Gas, Piston, Atmosphere → 3 seperate objects

$W_{gas} = PA\Delta x$, $W_{piston} = (P_{ambient} - P)A\Delta x$, $W_{atm} = -P_{amb}A\Delta x$

- work done in different processes:

a) isobaric: $P(V_2 - V_1)$ b) isothermal: $PV\ln(V_2/V_1)$

c) Polytrophic: $(P_2V_2 - P_1V_1)/(1-n)$ d) isochoric: 0

e) spring: $-\frac{1}{2}kx^2$ f) capacitor: $\frac{1}{2}CV^2$ g) resistor: i^2Rt $IVit$

h) gravity: $-mgh$ i) shaft: $2\pi N$(torque)(time)

Temperature and Heat:

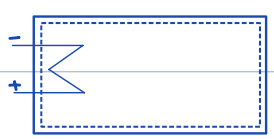
- When 2 bodies at different temperatures are brought into contact, after sometime they attain a common temperature → thermal equilibrium.
- Temperature may be understood as that property which determines whether 2 systems are in thermal eq. or not.
- Zeroth law of thermodynamics: If system A is in eq. with system B and system B is in eq. with system C, then systems A and C must be in equilibrium with each other.
- Thermometers: Const pressure gas thermometer, const volume gas thermometer, electrical resistance thermometer, Thermocouple.
- Length of 'Cu' rod changes when it is heated / cooled. The change in length is proportional to its temperature change.
- Ideally we require the property to change linearly with temperature.
- $°F = 1.8°C + 32$, $K = °C + 273.15$, $°R = 1.8K$, $°R, °K \geq 0$

- At 611.2 Pa, 0.01°C → water exists in 3 phases
- T = 273.16 K/K_T
- Heat: energy interaction between a system and surrounding that occus solely due to temperature diff between the two.
- Heat is energy in transit, Heat can't be stored
- Heat interaction is only defined at system boundary
- $Heat_{surrounding\ to\ system} \rightarrow +ve$, $Heat_{system\ to\ surrounding} \rightarrow -ve$
- In phase change situations, heat transferred to the system is utilized for heat required for phase change & temperature remains constant in those cases.
- If there's an temperature rise it may have not resulted from heat transfer (rubbing palms)
- Adiabatic system: No heat transfer.
- Modes of heat transfer:

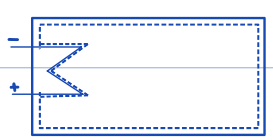
  a) Conduction $\dot{Q} = -KA\,dT/dx$ b) convection $\dot{Q} = hA(T_1 - T_2)$

  c) Radiation $\dot{Q} = \sigma \epsilon A(T_1^4 - T_2^4)$

-



Q = 0
W < 0



Q > 0
W = 0

First law for system:

- Work & Heat flow are path functions
- $\Sigma Q - \Sigma W = 0$ for cyclic process $\Rightarrow \oint (\delta Q - \delta W) = 0$

  $\Rightarrow \delta Q - \delta W = dE$ (E is a property)
- $Q - W = \Delta E$, E is stored energy

  $\Delta E = \Delta KE + \Delta PE + \Delta EE + \Delta ME + \Delta U$
- Only displacement work $\Rightarrow \Delta E = \Delta U$

  $Q - W = \Delta U$ or $\delta Q - \delta W = dU \Rightarrow \delta Q = dU + pdV$

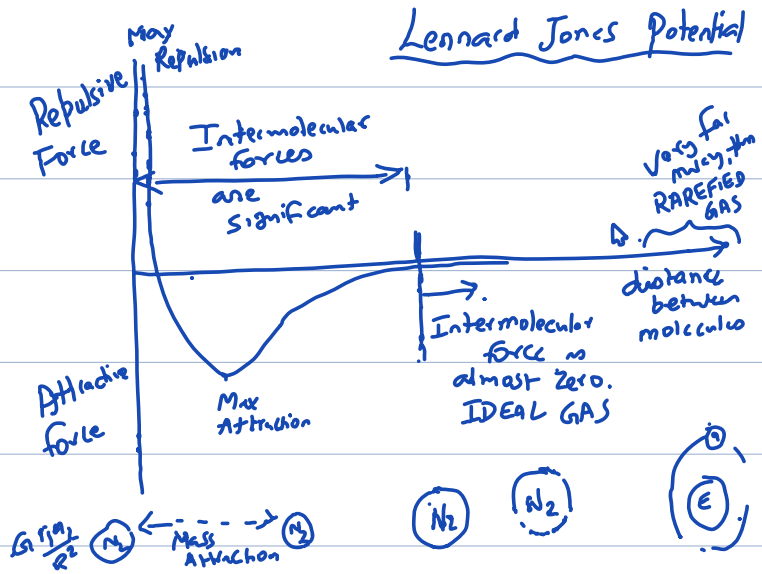
  infinitesimal heat added — internal energy change
- $H = U + PV$ (H is enthalpy) $\Rightarrow \delta Q = \delta H - V dP$
- $C_p = \left(\frac{\partial h}{\partial T}\right)_p$, $C_v = \left(\frac{\partial u}{\partial T}\right)_v$

Pure Substances:

- Substance which has uniform composition and chemical aggregation throughout the process under consideration. (air, water - steam mixture)
- Gibbs Phase rule: $F = C - \phi + 2$
- Two - Property rule: The state of a pure substance of

given mass can be specified in terms of two independent properties.

- In liquid-vapor system, specifying one quantity fixes the other ($\because F = 1$)
- Triple point $F = 0$
- **Lennard Jones Potential**



- $pV = n\bar{R}T = mRT$ $\quad \bar{R} = 8314 \frac{J}{kmol\,K}$ $\quad R = \frac{\bar{R}}{M} \rightarrow kg/kmol$
- In general for any pure substance u depends on 2 properties specifying the state $u = u(T)$ (→ through joule's experiment), $h = h(T)$

$$h = u + pv = u + RT$$

- Internal Energy of a gas depends on Translational, Rotational, Vibrational KE of molecules

- $U = \frac{1}{2} k_B T$ $\quad k_B = 1.38 \times 10^{-26}\ kJ/K = \frac{\bar{R}}{N_A}$ $\quad N_A = 6.023 \times 10^{26}\ (kmol)^{-1}$
- D → total degrees of freedom

$$u_{\text{per molecule}} = \frac{1}{2} k_B T D \Rightarrow \bar{u} = \frac{D}{2}\bar{R}T \quad \bar{h} = \frac{D+2}{2}\bar{R}T$$

$$\bar{c}_v = \frac{D}{2}\bar{R}, \quad \bar{c}_p = \frac{D+2}{2}\bar{R} \Rightarrow u = \frac{D}{2}RT \quad h = \frac{D+2}{2}RT$$

$$c_v = \frac{D}{2}R, \quad c_p = \frac{D+2}{2}R \Rightarrow \gamma = \frac{c_p}{c_v} = \frac{D+2}{D} \Rightarrow c_p = c_v + R$$

- D changes with temperature
- for perfect gases, specific heats are constant
- mixture of N constituents

$$x_i = n_i/n \qquad y_i = m_i/m \qquad y_i = \frac{x_i M_i}{m}, \quad x_i = \frac{y_i/M_i}{\sum_{j=1}^{N} y_j/M_j}$$

- Dalton's law of Partial Pressure: Each component in a mixture can be assumed to exert partial pressure at the mixture temperature and volume such that sum of all partial pressures is equal to the total pressure of the mixture. $\Rightarrow p_i/p = x_i$
- Amagat's Law: Each component in a mixture can be assumed to exert partial volume at the mixture temperature and pressure such that sum of all

partial volumes is equal to the total volume of the mixture. $\Rightarrow V_i/V = x_i$

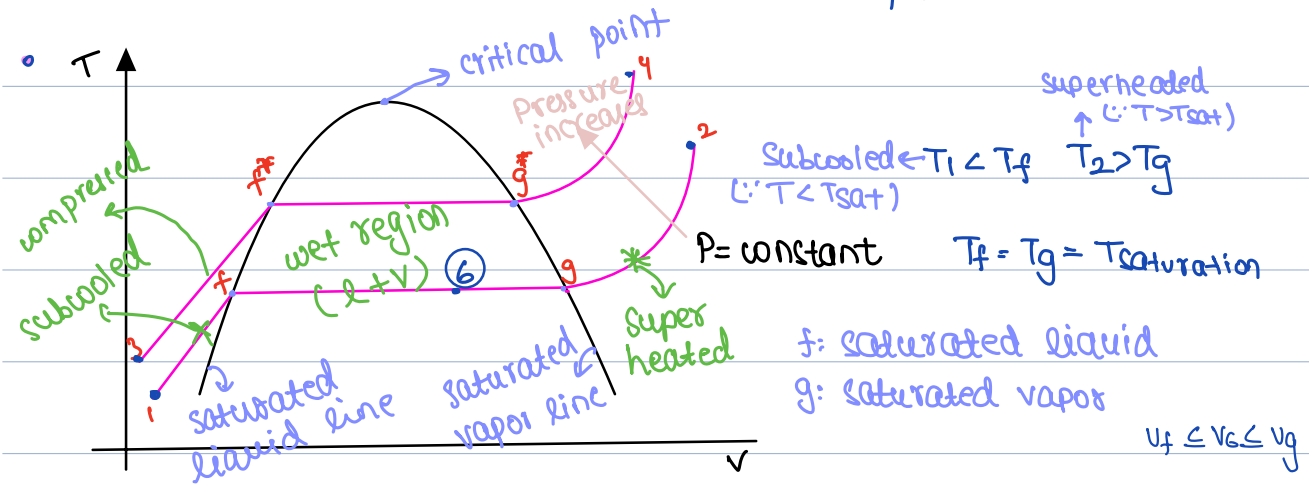
- $U = \Sigma m_i u_i \Rightarrow u = U/m = \Sigma y_i u_i \qquad \bar{u} = U/n = \Sigma x_i \bar{u}_i$

  $H = \Sigma m_i h_i \Rightarrow h = H/m = \Sigma y_i h_i \qquad \bar{H} = H/n = \Sigma x_i \bar{H}_i$

- $R_{mix} = \dfrac{\bar{R}}{M_{mix}} \qquad M_{mix} = \dfrac{m_1 + m_2 + \dots}{n_1 + n_2 + \dots}$

- Critical point: Specifies the conditions (T, P) above which distinct liquid and vapor phases cease to exist

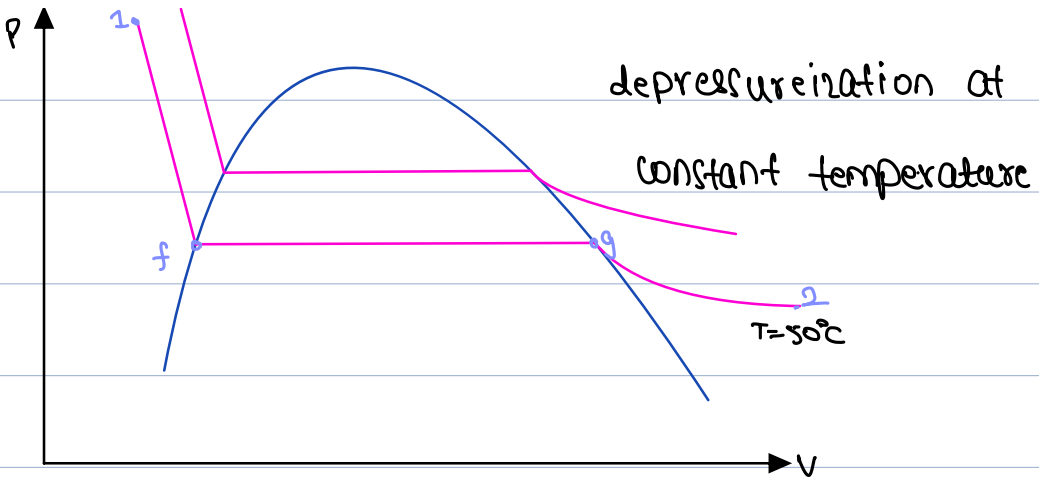
  For water $T_c = 647\,K$ and $P_c = 22\,MPa = 220\,bar$



$x = \dfrac{m_v}{m_v + m_l}$, $x_f = 0$, $x_g = 1 \Rightarrow$ quality (dryness fraction) $\dfrac{v_i - v_f}{v_g - v_f} = \dfrac{x_i - x_f}{x_g - x_f} = x_i$

for point 6

- water-steam mixture fails miserably as an ideal gas hence forget ideal gas laws.



- $v_i = x_i v_g + (1 - x_i) v_f$ (same of u, h)
  [interpolated value]

- If the condition pertains to sub-cooled liquid, then use temperature-based saturation table and write

  $h = u_f + pv_f$ (at temperature T_f)

  This value of enthalpy will usually be very close to h_f.

First law for flow processes:

- For cv the work interaction doesn't include disp work. It includes other forms of work (external)



$\frac{W_{in}}{m_{in}} = \frac{p_1 A_1 dx}{\rho_1 A_1 dx} = p_1 v_1 \rightarrow -ve$

$\frac{W_{out}}{m_{out}} = p_2 v_2 \rightarrow +ve$

Flow work: pv

- $m_{cv}(t) + dm_i = m_{cv}(t+\Delta t) + dm_e \Rightarrow \frac{d}{dt} m_{cv} = \dot{m}_i - \dot{m}_e$

  Steady state $\Rightarrow \frac{d}{dt} m_{cv} = 0$

- $\delta Q - (\delta W_{ext} + dm_e(pe\,ve) - dm_i(p_i v_i)) =$

  $E_{cv}(t+\Delta t) + dm_e\left(u_e + \frac{v_e^2}{2} + g z_e\right) - \left\{E_{cv}(t) + dm_i\left(u_i + \frac{v_i^2}{2} + g z_i\right)\right\}$

  $\Rightarrow \frac{dE_{cv}}{dt} = Q' - W'_{ext} + \dot{m}_i\left(h_i + \frac{v_i^2}{2} + g z_i\right) - \dot{m}_e\left(h_e + \frac{v_e^2}{2} + g z_e\right)$

  $E_{cv} = U_{cv} + KE_{cv} + PE_{cv}$

  Steady state $\Rightarrow \frac{d}{dt} E_{cv} = 0$

- **1st law for control volume**

  1. $\frac{dm_{cv}}{dt} = \Sigma \dot{m}_i - \Sigma \dot{m}_e$

  Steady state:
  m_{cv} = constant
  E_{cv} = constant

  2. $Q' - W'_{ext} = \frac{dE_{cv}}{dt} + \dot{m}_e\left(h_e + \frac{v_e^2}{2} + g z_e\right) - \dot{m}_i\left(h_i + \frac{v_i^2}{2} + g z_i\right)$

  3. $E_{cv} = U_{cv} + KE_{cv} + PE_{cv}$; $\frac{dE_{cv}}{dt} \approx \frac{dU_{cv}}{dt} = \frac{d}{dt}(m_{cv} u_{cv})$

- Each component of a power plant can be treated as a control volume and steady-state device.

- Turbine: $Q' = 0$, $\Delta PE = 0$, $\Delta KE = 0$

  power produced by turbine $W'_{ext} = \dot{m}(h_i - h_e) > 0$

- Heat exchanger: $\dot{W}'_{ext} = 0$, $\Delta PE = 0$, $\Delta KE = 0$

  rate of heat transfer $\dot{Q}' = \dot{m}(h_e - h_i)$

  If multiple inlet-outlets $\Rightarrow \dot{Q}' = 0 \Rightarrow \Sigma \dot{m}_e h_e = \Sigma \dot{m}_i h_i$

- Water pump: $\dot{Q}' = 0$, $\Delta KE = 0$, $\Delta PE = 0$, liquid is incompressible, no temperature change

  power required by pump $\dot{W}_p' = -\dot{W}_{ext}' = \dot{m}(h_e - h_i) > 0$

- Compressor: $\dot{Q}' = 0$, $\Delta PE = 0$, $\Delta KE = 0$

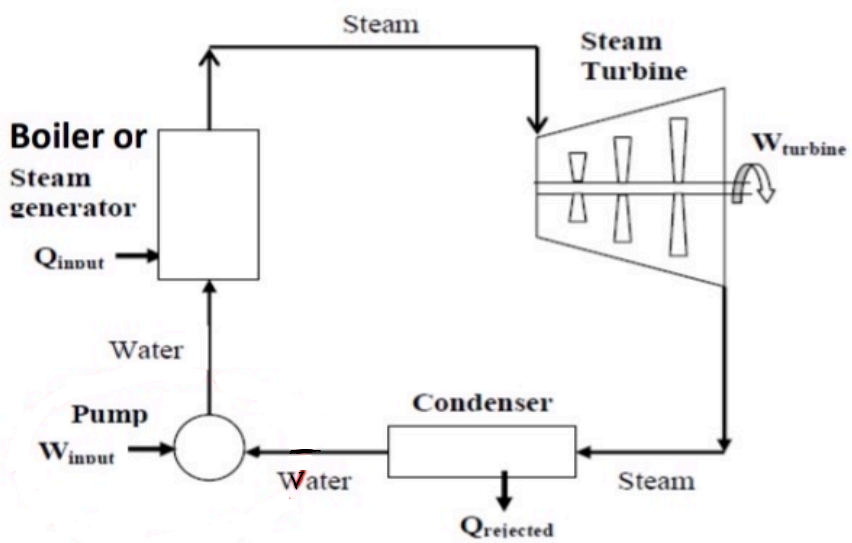
  Power needed by compressor $\dot{W}_c' = -\dot{W}_{ext}' = \dot{m}(h_e - h_i) > 0$

- Nozzle / Diffusor: $\dot{Q}' = 0$, $\dot{W}'_{ext} = 0$, $\Delta KE \neq 0$, $\Delta PE = 0$

  $\rho_i A_i v_i = \rho_e A_e v_e$. Generally in nozzle $v_i << v_e$ and diffusor $v_i >> v_e$. $h_e + \frac{v_e^2}{2} = h_i + \frac{v_i^2}{2} \Rightarrow (h_o)_e = (h_o)_i$

  $h_o \rightarrow$ Stagnation enthalpy

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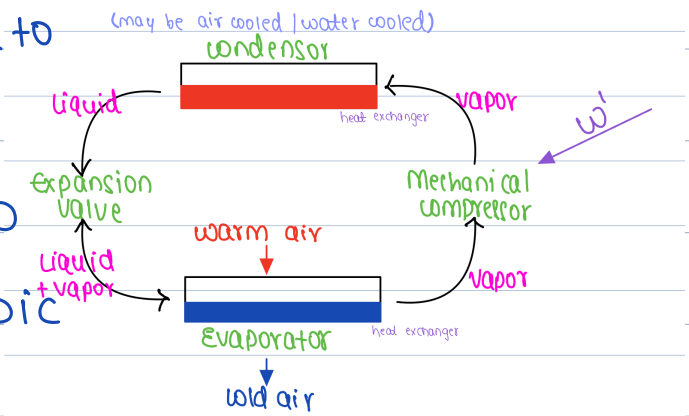


- Vapor compression refrigeration cycle:

  expansion valve: used to control flow rates.

  $\dot{Q} = \dot{W}_{ext} = \Delta PE = \Delta KE = 0$

  $\Rightarrow h_e = h_i$ ⟶ isenthalpic process

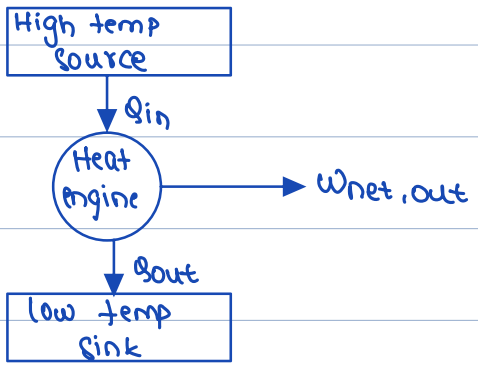


- $\dot{W}_{ext}$ for blower / fan < 0

Second Law:

- Second law: directionality and puts limits on conversion of energy from one form to another.
- Thermal reservoir: large body of matter that has ∞ heat capacity
- Source: Thermal reservoir that supplies heat
- Sink: Thermal reservoir that absorbs heat
- Heat engine: continuously operating device comprising a thermodynamic system operating in a thermodynamic cycle with heat & work interactions.
  ⟶ with source & sink

- For continuous operation, cyclic process is a must



- For analysis of heat engines, no details are required. Only temperatures of source & sink, quantity of heat transfer and work quantity is required.
- Direct engines: Heat to work (Power plants)
- Reverse engines: work to heat (refrigerators)
- The transfer of heat from a low temp medium to a high temp one requires special devices called refrigerators. The working fluid used in the refrigeration cycle is called a refrigerant.
- Thermal efficiency $(\eta) = \frac{\text{Net work output}}{\text{Total heat input}} = \frac{Q_H - Q_L}{Q_H}$

$$\eta = 1 - \frac{Q_L}{Q_H}$$

- $COP_{ref} = \frac{Q_L}{W_{in}}$, $COP_{HP} = \frac{Q_H}{W_{in}}$, $W_{in} = Q_H - Q_L$

- In theory is 100% work to heat possible? Yes
  more ordered (see piston cylinder example)
- In theory is 100% heat to work possible? No
  randomness increases as thermal energy rises
- Kelvin Plank statement: It is impossible to construct a device that will operate in a thermodynamic cycle and produce a net amount of work by exchange of heat with a single reservoir.
- Clausius statement: It is impossible to construct a device that operates in a thermodynamic cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body.
- Violation of Kelvin-Plank statement leads to the violation of Clausius statement also!
- Violation of Clausius statement leads to the violation of Kelvin-Plank statement too!
- Reversible process: process that retraces its path

if the cause is reversed, without leaving any trace in the surroundings (Quasi-static process)

- Reversible process delivers the maximum amount of work and consumes least amount of work.
- Irreversibilities occuring within the system boundary are called internal irreversibilities, and those outside the boundary are called external irreversibilities.
- The Carnot Principle: The efficiency of a irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs.
- The efficiencies of all reversible heat engines operating between the same two reservoirs are the same.
- The efficiency of a reversible heat engine is independent of
  - working fluid employed
  - the way the cycle is executed

- type of reversible engine used

- $\left(\frac{Q_H}{Q_L}\right)_{rev} = \frac{T_H}{T_L}$ (T in kelvin), $\eta_{th,rev} = 1 - \frac{T_L}{T_H}$

  $T_H, T_L \rightarrow$ working fluid temperatures

- 1) $\eta_{HE} = \frac{W_{net}}{Q_H} = 1 - \frac{Q_L}{Q_H}$ 3) $COP_{Ref} = \frac{Q_L}{W_{in}}$

  2) $\eta_{HE,R} = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_L}{T_H}$ 4) $COP_{HP} = \frac{Q_H}{W_{in}}$

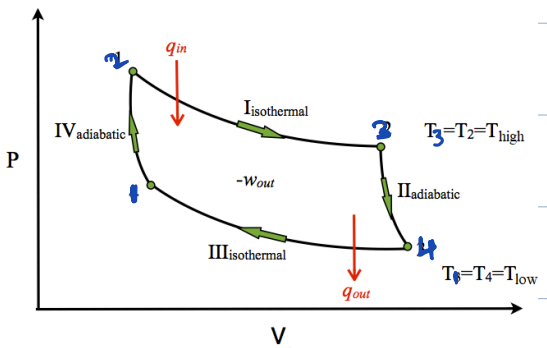
  5) $\eta_{HE,R} > \eta_{HE,I}$ 6) $COP_{Ref,R} > COP_{Ref,I}$

- Carnot's cycle : Thermodynamic cycle that avoids all irreversible processes. Simple carnot's cycle comprises reversible isothermal processes followed by reversible adiabatic processes.

- 2 rev adiabatic (isoentropic)
  - compression
  - expansion

  } W_{net} (>0) → Heat Engine

  2 rev isotherms
  - heat addition to working fluid $3 Q_H$
  - heat rejection from working fluid $3 Q_L$



T_H and T_L → temperature of reservoirs and working fluid

- η for carnot's engine:

process	Q	W	ΔU
1→2	0	$-mC_V(T_H-T_L)$	$mC_V(T_H-T_L)$
2→3	$mRT_H \ln(V_3/V_2)$	$mRT_H \ln(V_3/V_2)$	0
3→4	0	$mC_V(T_H-T_L)$	$-mC_V(T_H-T_L)$
4→1	$mRT_L \ln(V_1/V_4)$	$mRT_L \ln(V_1/V_4)$	0

$$W_{net} = mRT_H \ln\left(\frac{V_3}{V_2}\right) + mRT_L \ln\left(\frac{V_1}{V_4}\right)$$

$$Q_H = mRT_H \ln(V_3/V_2)$$

$$\eta = \frac{W_{net}}{Q_H} = 1 + \frac{T_L}{T_H}\frac{\ln(V_1/V_4)}{\ln(V_3/V_2)}$$

$$\frac{V_1}{V_2} = \frac{V_4}{V_3}$$

$$\frac{\ln(V_1/V_4)}{\ln(V_3/V_2)} = -1$$

$$\boxed{\eta = 1 - \frac{T_L}{T_H}}$$

Entropy:

- $\left(\frac{Q_H}{Q_L}\right)_{rev} = \frac{T_H}{T_L} \Rightarrow \oint \left(\frac{\delta Q}{T}\right)_{rev} = 0$

- $\frac{\delta Q_{rev}}{T} = dS =$ change in property $\Rightarrow S_2 - S_1 = \int_1^2 \frac{\delta Q_{rev}}{T}$

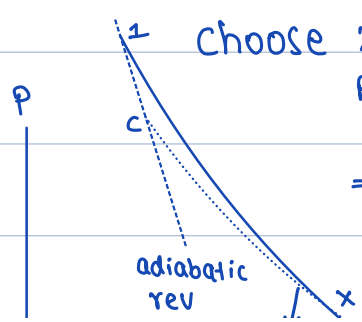
  $S \rightarrow$ entropy (kJ/K)

- Unlike energy, entropy is not conserved and thus there is no such concept as conservation of Entropy.
- Entropy is associated with the degree of randomness of system $\Rightarrow$ $S_{gas} > S_{liquid} > S_{solid}$

  Solid $\xrightarrow{heat}$ Liquid $\xrightarrow{heat}$ Gas $\Rightarrow dS = \frac{\delta Q_{rev}}{T} > 0$
- If End states are given, entropy change can be found easily, but we need not find heat transfer.
- There is no increase in entropy due to work transfer
- $dS = \frac{\delta Q_{rev}}{T}$ adiabatic (reversible) $\Rightarrow \delta Q_{rev} = 0 \Rightarrow dS = 0$

  $\Rightarrow$ isentropic process

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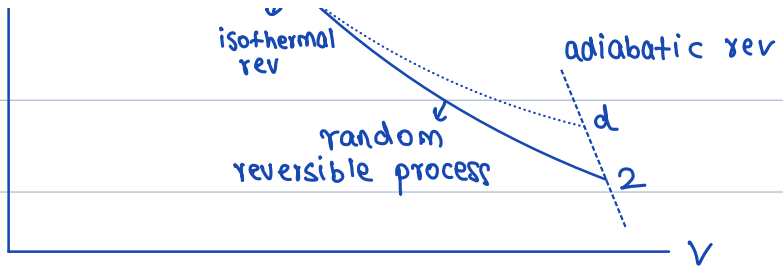


Choose x such that:

Area 1-c-x-1 = Area x-d-2-x

$\Rightarrow W_{1\to 2} = W_{1-c-d-2}$

$(\Delta U)_{1\to 2} = (\Delta U)_{1\to c\to d\to 2}$



Therefore $Q_{1\to 2} = Q_{1\to c\to d\to 2}$ (from first law)

- Every reversible process can be approximated by a combination of reversible isothermal and adiabatic processes.

- $\eta_{irr,HE} < \eta_{rev,HE} \Rightarrow (Q_L)_{rev} < (Q_L)_{irr}$ (Same Q_H)

$$\oint \left(\frac{\delta Q}{T}\right)_{irr} = \frac{Q_H}{T_H} - \frac{Q_{L,irr}}{T_L} < 0 \Rightarrow \text{In general } \oint \left(\frac{\delta Q}{T}\right) \leq 0$$

(Clausius inequality)

- $dS > \left(\frac{\delta Q}{T}\right)_{irr}$ or in general $dS \geq \left(\frac{\delta Q}{T}\right)$
- heat transfer to a system → increase in entropy
- work on/by system (reversible) will not change entropy
- work on/by system (including dissipative forces) will increase entropy
- Any irreversibility like heat transfer through a finite temperature difference, friction or other dissipative

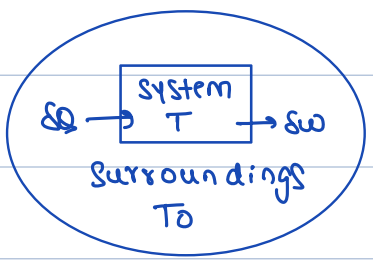
forces will only increase entropy.

$\Delta S_{system} = \Delta S_{\text{due to heat transfer}} + \delta S_{production}$

Entropy generated due to dissipative effect ≥ 0

- $dS = \frac{\delta Q}{T} + \delta S_{gen}$, $\delta S_{gen} \geq 0$
- $dS_{sys} \geq \frac{\delta Q}{T}$, $dS_{surr} = -\frac{\delta Q}{T_0} \Rightarrow dS_{net} = dS_{sys} + dS_{surr}$

$dS_{net} \geq \delta Q\left(\frac{1}{T} - \frac{1}{T_0}\right) > 0 \quad (\because T_0 > T)$



dS_{net} is total entropy generation in this process $\Rightarrow dS_{sys} + dS_{surr} > 0$

principle of increase in entropy

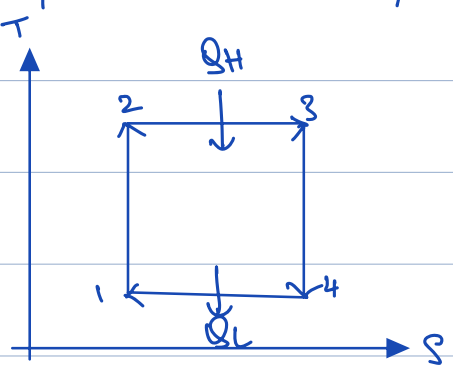
- $dS_{\text{isolated system}} \geq 0$ (equality if it's reversible processes only) (entropy can never decrease)
- $dS_{\text{adiabatic system}} \geq 0$
- Area under T-S curve gives Q_{rev}.
- The principle of increase in entropy indicates that All the processes that can take place will only increase the entropy of the system plus surroundings put together or in the limit keep the sum of their entropies constant. In other

words the system and the surroundings cannot be returned to their original states

- $Q_{rev} = \int_1^2 T\,dS$. If its a cycle then area enclosed by T-S graph indicates the net heat transfer which is also the net work done (reversible)
- Reversible adiabatic process $\Rightarrow Q = 0 = \delta S_{gen} \Rightarrow dS = 0$
- Only adiabatic process $\Rightarrow \delta Q = 0 \Rightarrow dS = \delta S_{gen} > 0$
- Irreversible processes should only be showed by dashed lines in graphs
- Carnot's cycle

  $W_{net} \rightarrow$ area under P-V

  $Q_{net} \rightarrow$ area under T-S



$$W_{net} = Q_{net} = (Q_H - Q_L) = Q_H + (-Q_L)$$

$$\eta = \frac{W_{net}}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

- Area under P-V curve will only represent work in a reversible process.
- Area under T-S curve will only represent heat transfer in a reversible process.

- $S_2 - S_1 = \int_1^2 dS + S_{gen}$

- $TdS = dU + pdV$

  $H = U + PV$

  $TdS = dH - VdP$

  $TdS = \delta Q_{rev}$

  valid for any process (both rev and irrev)

$TdS = mC_v dT + \frac{mRT}{V} dV \Rightarrow dS = mC_v \frac{dT}{T} + mR \frac{dV}{V}$

$\Rightarrow \boxed{\Delta S = mC_v \ln\left(\frac{T_2}{T_1}\right) + mR \ln(V_2/V_1)}$

calorifically perfect gas: C_v isn't function of temperature

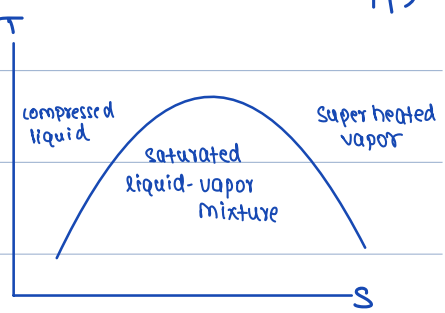
$\boxed{\Delta S = mC_p \ln(T_2/T_1) - mR \ln(P_2/P_1)}$

- In solids and liquids, when temperature changes volume change is negligible. $\Rightarrow C_p = C_v = C$

  $TdS = dU + \cancel{pdV}^{0} \Rightarrow TdS = dU = mCdT \Rightarrow \Delta S = mC \ln\left(\frac{T_2}{T_1}\right)$

- In Steam table: $S = (1-x) S_f + x S_g$

  $S_{compliq}(T, P) = S_{sat,liq}(T)$



- For irreversible system,

  $\delta W_{irr} \neq pdV$, $\delta Q_{irr} \neq TdS \Rightarrow \delta Q_{irr} = TdS - T\delta S_{gen}$

  $\delta Q_{irr} = dU + \delta W_{irr} \Rightarrow \delta W_{irr} = pdV - T\delta S_{gen}$

  $T\delta S_{gen}$ is often called lost work

- Turbine: shaft work is produced at expense of enthalpy of fluid that flows through it, generally adiabatic, ΔKE generally 0. $\frac{dW_{out}}{dt} = \frac{dm}{dt}(h_i - h_e)$. (>0)

η (isentropic efficiency) $= \frac{W_{actual}}{W_{isentropic}} = \frac{C_p(T_i - T_{ea})}{C_p(T_i - T_{es})}$

- Compressor: uses shaft work input to increase pressure of flowing fluid, flow occurs from lower pressure to higher pressure, adiabatic generally, ΔKE generally 0. $\frac{dW_{in}}{dt} = \frac{dm}{dt}(h_e - h_i)$ (>0)

η (isentropic efficiency) $= \frac{W_{isentropic}}{W_{actual}} = \frac{C_p(T_{es} - T_i)}{C_p(T_{ea} - T_i)}$

- Nozzle: Associated with no shaft work, flow occurs from higher pressure to lower pressure, generally adiabatic, Generally KE_{inlet} is negligible.

$\frac{v_e^2}{2} - \frac{v_i^2}{2} = h_i - h_e$.

η (isentropic efficiency) $= \frac{V_{actual}^2/2}{V_{isentropic}^2/2} = \frac{C_p(T_i - T_{ea})}{C_p(T_i - T_{es})}$